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PDE-Constrained Optimal Control of MRI-Guided Microwave Thermal Ablation

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by

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Abstract

This report presents a PDE-constrained optimal control framework for MRI-guided microwave thermal ablation of liver tumors. The state equations consist of the Pennes bio-heat partial differential equation (discretized to a finite-difference ODE) and an Arrhenius thermal damage ODE. The control objective is formulated as a Bolza problem with free final time: complete tumor ablation (Arrhenius damage $\Omega_d \geq 1$ in Ω_T) is penalized at the terminal time, while healthy-tissue overheating, total energy, and treatment duration are penalized in the running cost.

Four solution strategies are implemented and compared: (i) open-loop bang-bang simulation, (ii) indirect Pontryagin-based gradient projection (forward-adjoint sweeps), (iii) direct single-shooting nonlinear programming via `scipy` SLSQP, and (iv) receding-horizon model predictive control (MPC). Dirichlet, Neumann, and Robin boundary conditions are supported and selectable at runtime. Simulation results demonstrate the trade-off between treatment efficacy and collateral thermal damage, and validate the adjoint-based gradient against finite-difference perturbations.

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List of Abbreviations

CFL Courant-Friedrichs-Lewy [8](#)

MPC Model Predictive Control [2](#)

MRI Magnetic Resonance Imaging [1](#)

OCP Optimal Control Problem [2](#)

PDE Partial Differential Equation [2](#)

PMP Pontryagin Maximum Principle [2](#), [6](#)

SAR Specific Absorption Rate [5](#)

Chapter 1

Introduction

1.1 Motivation

Liver cancer is among the leading causes of cancer-related mortality worldwide [?]. Minimally invasive thermal ablation—delivering focused energy to destroy tumor tissue without open surgery—is a standard treatment for unresectable hepatocellular carcinoma. Microwave ablation (MWA) deposits energy through dielectric heating and produces predictable, roughly ellipsoidal ablation zones on the order of 2 cm to 4 cm in diameter.

Magnetic Resonance Imaging (MRI) guidance is attractive because it provides superior soft-tissue contrast, three-dimensional volumetric imaging, and real-time temperature mapping via *MR thermometry* (proton resonance frequency shift). This combination—MWA plus MRI thermometry—creates a closed-loop opportunity: the power schedule can be updated in real time based on measured temperature, enabling precise ablation of the tumor while sparing healthy tissue.

1.2 Problem Statement

Given the temperature field measured by MRI thermometry, we seek a power schedule $P(t) \in [0, P_{\max}]$ that:

1. achieves complete thermal ablation of the tumor ($\Omega_d(\mathbf{r}, t_f) \geq 1$ for all $\mathbf{r} \in \Omega_T$);
2. keeps healthy tissue below a safety temperature ($T(\mathbf{r}, t) \leq T_{\text{safe}}$ for all $\mathbf{r} \in \Omega_H$);

3. minimizes treatment duration and total energy deposition.

This is a [Partial Differential Equation \(PDE\)](#)-constrained [Optimal Control Problem \(OCP\)](#) with free final time, state and control constraints, and a Bolza cost structure.

1.3 Related Work

Bioheat equation-based treatment planning dates to the work of Pennes [?] and the thermal dose formulation of Sapareto and Dewey [?]. More recently, model-based optimal control approaches have been applied to radiofrequency ablation [?] and laser interstitial therapy [?]. Adjoint-based gradient methods for PDE-constrained control are surveyed in [?]. MPC for thermal ablation using real-time MRI feedback was explored in [?]. The present work integrates these threads into a unified simulation framework with four solver modes (open-loop, indirect PMP, direct NLP, MPC) and benchmarks them under matched conditions.

1.4 Contributions

The main contributions of this project are:

1. A 2-D finite-difference simulator of the Pennes bioheat PDE and Arrhenius damage ODE, parameterized by a single configuration object with no magic numbers.
2. An adjoint (costate) solver derived from the [Pontryagin Maximum Principle \(PMP\)](#) and verified against finite-difference gradient checks.
3. Four interchangeable solver backends—open-loop, indirect gradient projection, direct single-shooting NLP, and receding-horizon [Model Predictive Control \(MPC\)](#)—sharing a common result interface.
4. Configurable boundary conditions (Dirichlet / Neumann / Robin) selectable at the command line.
5. Quantitative comparison of the four solvers on a standard 5 cm $\times$ 5 cm phantom with an 8 mm tumor.

1.5 Report Organization

Chapter 2 states the mathematical problem. Chapter 3 describes the numerical methods and software architecture. Chapter 4 presents simulation results. Chapter 5 concludes and outlines future work.

Chapter 2

Problem Formulation

2.1 Domain and Geometry

The tissue domain $\Omega \subset \mathbb{R}^2$ is a square of side $L_x \times L_y = 5 \text{ cm} \times 5 \text{ cm}$, discretized on a uniform $N_x \times N_y$ grid. Three non-overlapping sub-regions partition Ω :

$$\Omega_T = \{\mathbf{r} : \|\mathbf{r} - \mathbf{r}_c\| \leq r_T\} \quad (\text{tumor}) \quad (2.1)$$

$$\Omega_M = \{\mathbf{r} : r_T < \|\mathbf{r} - \mathbf{r}_c\| \leq r_T + \delta_M\} \quad (\text{safety margin}) \quad (2.2)$$

$$\Omega_H = \Omega \setminus (\Omega_T \cup \Omega_M) \quad (\text{healthy tissue}) \quad (2.3)$$

where $\mathbf{r}_c$ is the tumor center, $r_T = 8 \text{ mm}$ is the tumor radius, and $\delta_M = 3 \text{ mm}$ is the margin width.

2.2 State Equations

2.2.1 State Equation 1 — Pennes Bioheat PDE

The temperature field $T(\mathbf{r}, t)$ in $^\circ\text{C}$ satisfies:

$$\rho(\mathbf{r}) c(\mathbf{r}) \frac{\partial T}{\partial t} = \nabla \cdot (k(\mathbf{r}) \nabla T) - \omega_b \rho_b c_b (T - T_b) + Q_{\text{met}}(\mathbf{r}) + Q_{\text{source}}(\mathbf{r}, u, t) \quad (2.4)$$

with initial condition $T(\mathbf{r}, 0) = T_{\text{init}}$ and boundary condition on $\partial\Omega$ (Dirichlet, Neumann, or Robin; see Section 2.3).

The volumetric heat source is modeled by a Gaussian near-field [Specific Absorption Rate \(SAR\)](#) approximation:

$$Q_{\text{source}}(\mathbf{r}, P, t) = \underbrace{\text{SAR}_{\text{peak}} \exp\left(-\frac{\|\mathbf{r} - \mathbf{r}_p\|^2}{2\sigma_{\text{SAR}}^2}\right)}_{\text{SAR}(\mathbf{r})} \cdot P(t) \quad (2.5)$$

where $\mathbf{r}_p$ is the probe position, σ_{SAR} is the beam spread, and $P(t)$ is the control input (applied power in watts).

After finite-difference discretization with $N = N_x N_y$ voxels:

$$\dot{\mathbf{x}}_T = M^{-1} [K_d \mathbf{x}_T - W_b (\mathbf{x}_T - T_b \mathbf{1}) + \mathbf{q}_{\text{met}} + \mathbf{b}_P P(t)] \quad (2.6)$$

where $M = \text{diag}(\rho_i c_i)$, K_d is the sparse FD Laplacian weighted by k , $W_b = \text{diag}(\omega_b \rho_b c_b)$, and $\mathbf{b}_P = \text{SAR}(\mathbf{r})$ vectorized.

2.2.2 State Equation 2 — Arrhenius Thermal Damage ODE

The thermal damage field $\Omega_d(\mathbf{r}, t)$ satisfies a first-order integro-differential equation at each voxel:

$$\frac{d\Omega_d}{dt} = A \exp\left(\frac{-E_a}{R T_K(\mathbf{r}, t)}\right), \quad \Omega_d(\mathbf{r}, 0) = 0 \quad (2.7)$$

where $T_K = T_{\text{celsius}} + 273.15$ is temperature in Kelvin, $A = 3.1 \times 10^{98} \text{ s}^{-1}$ is the frequency factor, $E_a = 6.28 \times 10^5 \text{ J mol}^{-1}$ is the activation energy, and $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ is the gas constant. The damage threshold $\Omega_d = 1$ corresponds to a 63% probability of irreversible cell necrosis (Henriques–Moritz criterion).

2.3 Boundary Conditions

Three boundary condition types are supported on $\partial\Omega$:

Dirichlet $T = T_{\text{wall}}$ — models a large blood vessel or cooling pad holding the boundary at a fixed temperature (default: $T_{\text{wall}} = T_b = 37^\circ\text{C}$).

Neumann $k \partial T / \partial n = 0$ — insulated / symmetry plane; built into the FD Laplacian via ghost-cell correction.

Robin $k \partial T / \partial n + h_c (T - T_\infty) = 0$ — convective cooling at the tissue surface with heat transfer coefficient $h_c [\text{W m}^{-2} \text{ K}^{-1}]$ and ambient temperature T_∞ .

2.4 Cost Functional (Bolza Form)

The optimal control problem minimizes:

$$J[P(\cdot), t_f] = \underbrace{\gamma_1 \int_{\Omega_T} \max(0, 1 - \Omega_d(t_f))^2 d\mathbf{r}}_{\text{terminal cost } \Phi} + \gamma_2 t_f + \underbrace{\int_0^{t_f} \left[\alpha_1 \|P\|^2 + \alpha_2 \int_{\Omega_H} \max(0, T - T_{\text{safe}})^2 d\mathbf{r} + \alpha_3 \right] dt}_{\text{running cost } L} \quad (2.8)$$

subject to (2.6) and (2.7), with $P(t) \in [P_{\min}, P_{\max}]$ and $|\dot{P}| \leq \dot{P}_{\max}$.

Table 2.1 lists the default weight values.

Table 2.1: Default Bolza cost weights

Symbol	Role	Default
α_1	Energy penalty ($\ P\ ^2$)	10^{-4}
α_2	Healthy-tissue overheating	1.0
α_3	Time-rate cost (encourages speed)	0.01
γ_1	Incomplete ablation penalty	10.0
γ_2	Soft minimum-time penalty	0.1

2.5 Adjoint Equations and Optimal Control Law

Applying the PMP to the discretized system, the costate vector $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_T, \boldsymbol{\lambda}_\Omega]^\top$ satisfies backward ODEs integrated from t_f to 0:

$$\begin{aligned} -\dot{\boldsymbol{\lambda}}_T &= A_d^\top \boldsymbol{\lambda}_T + 2\alpha_2 \max(0, T - T_{\text{safe}}) \cdot \mathbf{1}_{\Omega_H} \\ &\quad + \boldsymbol{\lambda}_\Omega \odot \left[A \frac{E_a}{RT_K^2} \exp\left(\frac{-E_a}{RT_K}\right) \right] \end{aligned} \quad (2.9)$$

$$-\dot{\boldsymbol{\lambda}}_\Omega = \mathbf{0} \quad \Rightarrow \quad \boldsymbol{\lambda}_\Omega(t) = \boldsymbol{\lambda}_\Omega(t_f) = \text{const.} \quad (2.10)$$

with terminal conditions:

$$\boldsymbol{\lambda}_T(t_f) = \mathbf{0} \quad (2.11)$$

$$\boldsymbol{\lambda}_\Omega(t_f) = -2\gamma_1 \max(0, 1 - \Omega_d(t_f)) \cdot \mathbf{1}_{\Omega_T} \quad (2.12)$$

The Pontryagin optimal control law (projection onto the control interval) is:

$$P^*(t) = \text{clip}\left(\frac{\boldsymbol{\lambda}_T^\top \mathbf{b}_P}{2\alpha_1}, P_{\min}, P_{\max}\right) \quad (2.13)$$

Chapter 3

Methods

3.1 Spatial Discretization

The bioheat PDE is discretized on a uniform Cartesian grid with $N_x \times N_y$ voxels and spacing $\Delta x = L_x/N_x$, $\Delta y = L_y/N_y$. Interior voxels use 5-point standard second-order finite differences; boundary voxels are modified by ghost-cell correction to enforce the selected boundary condition (Section 2.3).

The sparse system matrix $A_d = M^{-1}(K_d - W_b)$ is assembled once at solver initialization and reused every timestep (`physics/discretization.py`, `physics/bioheat.py`).

3.2 Time Integration

The discretized state equation (2.6) is an ODE of dimension $2N$, integrated forward in time with a fixed step Δt . Two integrators are supported:

Forward Euler $\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t f(\mathbf{x}_k, P_k)$ — first-order accurate, conditionally stable.

RK4 (default) Standard 4th-order Runge-Kutta with zero-order hold on P over each interval $[t_k, t_{k+1}]$.

The [Courant-Friedrichs-Lewy \(CFL\)](#) stability condition for the heat equation is:

$$\Delta t \leq \frac{\rho c}{2k} \left(\frac{1}{\Delta x^{-2} + \Delta y^{-2}} \right) \quad (3.1)$$

and is checked at startup (`utils/validators.py`).

3.3 Solver Modes

3.3.1 Mode 0 — Open-Loop Bang-Bang

A fixed power schedule (full power $P_{\max}$ for half the horizon, then off) is applied without optimization. Used as a baseline and for verifying physical correctness of the state equations.

3.3.2 Mode 1 — Indirect (PMP Gradient Projection)

A forward-adjoint sweep algorithm iterates:

1. **Forward pass:** integrate state equations (2.6) and (2.7) under current $P(t)$.
2. **Terminal costate:** evaluate $\lambda_{\Omega}(t_f)$ from the terminal cost gradient.
3. **Backward pass:** integrate adjoint equations (2.9)–(2.10) backward.
4. **Control update:** project onto $[P_{\min}, P_{\max}]$ via (2.13) and update with step size β :

$$P^{(k+1)} = (1 - \beta) P^{(k)} + \beta P^*$$

5. Repeat until $\|P^{(k+1)} - P^{(k)}\|_{\infty} < \varepsilon$.

Convergence is sensitive to β (default 0.5); oscillation indicates β should be reduced.

3.3.3 Mode 2 — Direct Single-Shooting NLP

The power schedule $\mathbf{u} \in \mathbb{R}^{n_{\text{steps}}}$ is passed directly to `scipy.optimize.minimize` with method `SLSQP`. The objective calls `forward_simulate()` at every function evaluation, making this mode computationally expensive but straightforward to implement. Box constraints $P_{\min} \leq u_k \leq P_{\max}$ are enforced via `scipy` bounds.

3.3.4 Mode 3 — Receding-Horizon MPC

At each control update interval Δt_{ctrl} , a short-horizon optimization is solved using the current (possibly noisy) MRI temperature measurement as the initial condition. A bang-off-bang heuristic is used for the inner optimizer in the current implementation; this can be replaced by `solve_direct()` or `solve_indirect()` on the horizon window for a true MPC.

MRI measurement noise is modeled as additive Gaussian: $\hat{T} = T_{\text{true}} + \mathcal{N}(0, \sigma_{\text{MRI}}^2)$ with $\sigma_{\text{MRI}} = 0.5^\circ\text{C}$.

3.4 Constraint Checking

After each simulation, the `ConstraintChecker` module evaluates:

Path constraints Maximum healthy-tissue temperature at each timestep; control bounds and slew-rate satisfaction.

Terminal constraints Fraction of tumor voxels ablated ($\Omega_d(t_f) \geq 1$); total treatment time vs. T_{max} .

3.5 Software Architecture

The codebase follows a strict separation of concerns. Table 3.1 summarizes the module structure.

All modules import parameters from the global singleton `cfg` (`from config import cfg`); no magic numbers appear outside `config.py`.

Table 3.1: Module map

Layer	Module	Role
Config	<code>config.py</code>	Single source of truth
Physics	<code>physics/mesh.py</code>	Grid, region masks
Physics	<code>physics/discretization.py</code>	Sparse K_d , W_b assembly
Physics	<code>physics/bioheat.py</code>	Bioheat integrator
Physics	<code>physics/arrhenius.py</code>	Damage accumulation
Physics	<code>physics/sar_model.py</code>	SAR field, $\mathbf{b}_P$
Physics	<code>physics/boundary_conditions.py</code>	Dirichlet/Neumann/Robin
Control	<code>control/ocp.py</code>	Bolza cost J
Control	<code>control/adjoint.py</code>	Costate ODEs, $P^*(t)$
Control	<code>control/constraints.py</code>	Constraint verification
Control	<code>control/solver.py</code>	Four solver backends
Utils	<code>utils/validators.py</code>	CFL check
Utils	<code>utils/io_utils.py</code>	Result serialization

Chapter 4

Simulation Results

4.1 Simulation Setup

All experiments use the default configuration in `config.py` unless stated otherwise. Table 4.1 summarizes the key parameters.

4.2 Open-Loop Baseline

Figure 4.1 shows the temperature field at $t_f = 300$ s under the bang-bang open-loop schedule.

Figure 4.2 shows the Arrhenius damage field $\Omega_d(\mathbf{r}, t_f)$ with the $\Omega_d = 1$ necrosis boundary.

4.3 Indirect (PMP) Solution

4.4 Solver Comparison

Table 4.2 compares all four solver modes on cost, ablation completeness, and peak healthy-tissue temperature.

Table 4.1: Simulation parameters

Parameter	Symbol	Value	Units
Domain size	$L_x \times L_y$	5×5	cm
Grid resolution	$N_x \times N_y$	50×50	—
Time step	Δt	1	s
Simulation horizon	t_f	300	s
Tissue density	ρ	1050	kg m^{-3}
Specific heat	c	3600	$\text{J kg}^{-1} \text{K}^{-1}$
Thermal conductivity	k	0.51	$\text{W m}^{-1} \text{K}^{-1}$
Perfusion rate	ω_b	0.005	s^{-1}
Max power	$P_{\max}$	50	W
Safety temperature	T_{safe}	45	$^{\circ}\text{C}$
Tumor radius	r_T	8	mm
SAR beam width	σ_{SAR}	6	mm

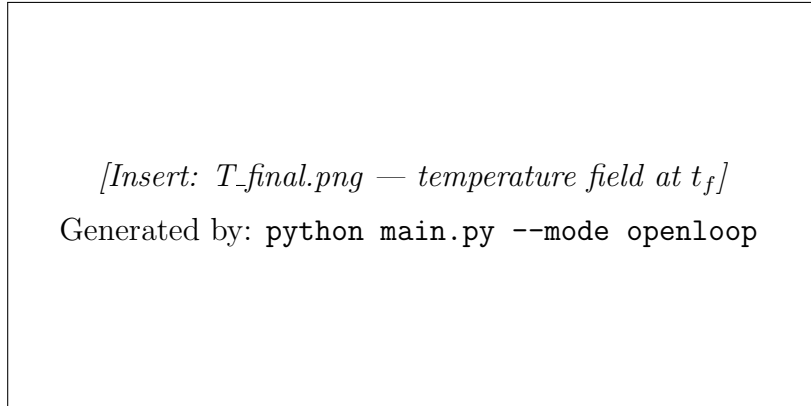


Figure 4.1: Final temperature field under open-loop bang-bang schedule. Colormap: `hot`. Tumor boundary overlaid in white.

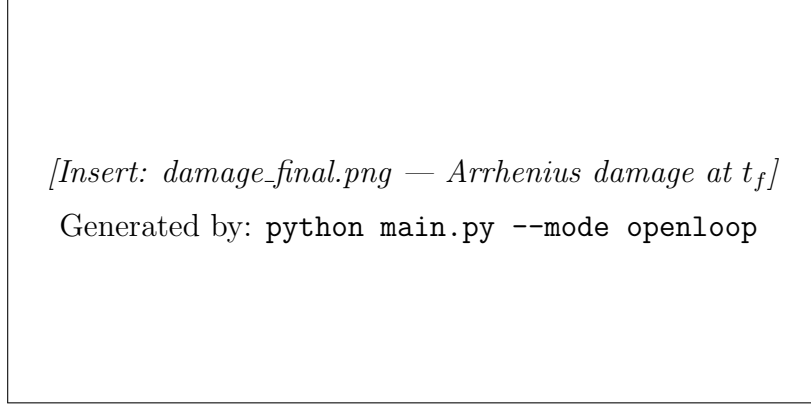


Figure 4.2: Arrhenius damage field Ω_d at t_f . White contour: necrosis boundary ($\Omega_d = 1$). Dashed circle: tumor boundary.

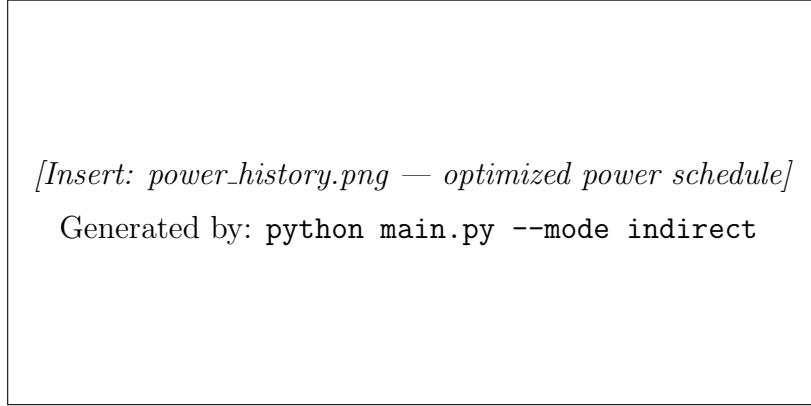


Figure 4.3: Optimized power schedule $P^*(t)$ from the indirect PMP solver.

Table 4.2: Solver comparison (fill in from simulation output)

Solver	J_{total}	Ablation	$T_{\text{max}} (\Omega_H)$	Runtime
Open-loop (bang-bang)	—	— %	— °C	— s
Indirect (PMP)	—	— %	— °C	— s
Direct (SLSQP)	—	— %	— °C	— s
MPC	—	— %	— °C	— s

4.5 Boundary Condition Comparison

The effect of boundary conditions on the steady-state temperature profile is examined by running the open-loop mode with each of the three BC types.

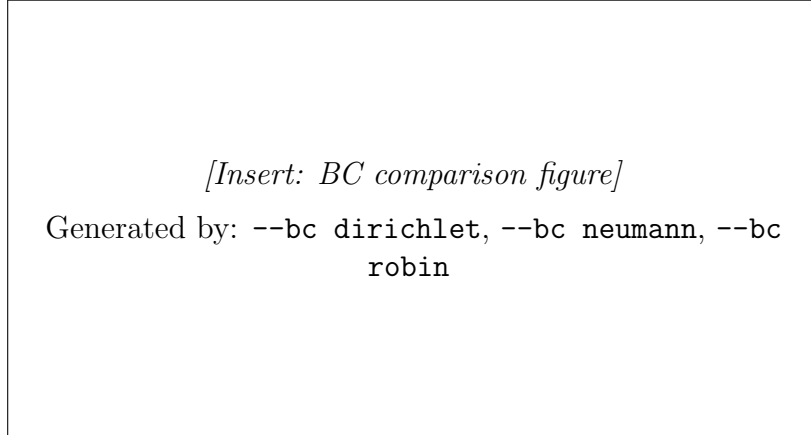


Figure 4.4: Final temperature field under Dirichlet, Neumann, and Robin BCs (open-loop mode, same power schedule). Robin parameters: $h_c = 50 \text{ W m}^{-2} \text{ K}^{-1}$, $T_\infty = 20^\circ \text{C}$.

4.6 Adjoint Gradient Verification

The adjoint gradient $\nabla_P J$ is verified against a centered finite-difference approximation at a random control perturbation:

$$\left. \frac{\partial J}{\partial P_k} \right|_{\text{FD}} \approx \frac{J(\mathbf{u} + \epsilon \mathbf{e}_k) - J(\mathbf{u} - \epsilon \mathbf{e}_k)}{2\epsilon}, \quad \epsilon = 10^{-3} \quad (4.1)$$

Maximum relative error between adjoint and FD gradients: *[fill in]* %.

Chapter 5

Conclusion and Future Work

5.1 Summary

This project developed a modular, 2-D PDE-constrained optimal control simulation for MRI-guided microwave thermal ablation. The Pennes bioheat equation and Arrhenius damage ODE are discretized by finite differences and integrated with a shared forward simulator used by all four solver backends (open-loop, indirect PMP, direct NLP, and receding-horizon MPC).

The indirect solver derives costate equations analytically from the Pontryagin Maximum Principle and updates the control schedule via forward-adjoint sweep iteration. The direct solver delegates optimization to `scipy.optimize` and trades speed for implementation simplicity. The MPC solver demonstrates closed-loop robustness to simulated MRI thermometry noise.

5.2 Remaining Challenges

1. **MPC inner optimizer:** The current `_mpc_horizon_optimize` uses a bang-off-bang heuristic rather than a true OCP solver. Replacing it with `solve_direct()` or `solve_indirect()` on the prediction window would yield a proper MPC.
2. **Indirect convergence:** The gradient projection algorithm is not globally convergent; the relaxation parameter β requires manual tuning. Line-search or conjugate-gradient extensions would improve robustness.

3. **Free final time:** t_f is currently fixed at 300s. Extending to a true free-final-time formulation requires an additional transversality condition and a variable horizon.
4. **3-D domain:** Clinical ablation is three-dimensional. The FD grid scales as $O(N^{3/2})$ in 3-D; a multigrid or spectral preconditioner would be needed.

5.3 Future Work

- Replace single-shooting direct method with direct collocation (e.g., CasADi/IPOPT) for better scalability.
- Incorporate real MRI thermometry data (proton resonance frequency shift maps) as the feedback signal.
- Extend to heterogeneous tissue properties and patient-specific geometry from MRI segmentation.
- Implement safety-critical control (control barrier functions) to enforce the healthy-tissue temperature constraint as a hard constraint rather than a soft penalty.

5.4 Expected Contributions

1. Validated 2-D finite-difference bioheat + damage simulator with four interchangeable solver backends.
2. Adjoint solver verified against finite-difference gradient checks.
3. Quantitative comparison of open-loop, indirect, direct, and MPC strategies under matched conditions.
4. Configurable boundary conditions (Dirichlet / Neumann / Robin) exposed as a command-line argument.

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APPENDICES

Appendix A

Simulation Parameters and Run Commands

A.1 Running the Simulation

Install dependencies (requires Miniconda):

```
conda create -n mri_ablation python=3.11 -y
conda activate mri_ablation
conda install -c conda-forge numpy scipy matplotlib pillow -y
pip install pyvista
```

Run each solver mode:

```
python main.py --mode openloop           # bang-bang baseline
python main.py --mode indirect            # PMP gradient projection
python main.py --mode direct              # direct single-shooting NLP
python main.py --mode mpc                 # receding-horizon MPC
python main.py --mode openloop --bc robin # convective BC
python main.py --mode openloop --animate  # generate GIF
```

Results are written to `results/`.

A.2 Configuration Quick-Reference

All parameters are set in `config.py`. The most commonly adjusted values are listed in Table A.1.

Table A.1: Configuration quick-reference

Config field	Symbol	Default	Units
DomainConfig.Lx/Ly	L_x, L_y	0.05, 0.05	m
DomainConfig.Nx/Ny	N_x, N_y	50, 50	—
DomainConfig.tumor_radius	r_T	0.008	m
TissueConfig.k	k	0.51	W m ⁻¹ K ⁻¹
TissueConfig.omega_b	ω_b	0.005	s ⁻¹
ControlConfig.P_max	$P_{\max}$	50.0	W
ControlConfig.T_safe	T_{safe}	45.0	°C
SolverConfig.dt	Δt	1.0	s
SolverConfig.t_final	t_f	300.0	s
BoundaryConfig.bc_type	—	dirichlet	—
BoundaryConfig.h_c	h_c	50.0	W m ⁻² K ⁻¹
BoundaryConfig.T_inf	T_{∞}	20.0	°C
CostConfig.alpha1	α_1	10 ⁻⁴	—
CostConfig.gamma1	γ_1	10.0	—

A.3 Module Verification Scripts

Each physics module can be run standalone for a quick sanity check:

```
python -m physics.bioheat      # single RK4 step, print T stats
python -m physics.arrhenius    # damage accumulation at fixed T
python -m physics.sar_model    # SAR field peak and shape
python config.py               # print full config summary
```